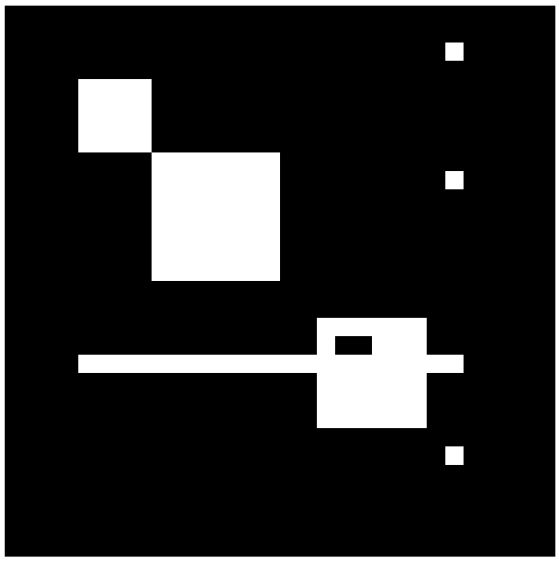
Binary Image Processing

```
img = false(30, 30);
img(5:8, 5:8) = true;
img(9:15, 9:15) = true;
img(18:23, 18:23) = true;
img(19:20, 19:20) = false;
img(5:8, 5:8) = true;
img(20, 5:25) = true;
img(3, 25) = true;
img(10, 25) = true;
img(25, 25) = true;
imshow(img)
```



STREL

```
getnhood(strel("square", 3))
```

```
ans = 3×3 logical array
   1   1   1
   1   1   1
   1   1   1
```

```
getnhood(strel("rectangle", [3 5]))
```

```
ans = 3×5 logical array
   1   1   1   1   1
   1   1   1   1   1
   1   1   1   1   1
```

```
getnhood(strel("diamond", 2))
```

```
ans = 5×5 logical array
   0   0   1   0   0
   0   1   1   1   0
   1   1   1   1   1
   0   1   1   1   0
   0   0   1   0   0
```

```
getnhood(strel("disk", 2))
```

```
ans = 5×5 logical array
   0   0   1   0   0
   0   1   1   1   0
   1   1   1   1   1
   0   1   1   1   0
   0   0   1   0   0
```

```
getnhood(strel("octagon", 3))
```

```
ans = 7×7 logical array
   0   0   1   1   1   0   0
   0   1   1   1   1   1   0
   1   1   1   1   1   1   1
   1   1   1   1   1   1   1
   1   1   1   1   1   1   1
   0   1   1   1   1   1   0
   0   0   1   1   1   0   0
```

```
getnhood(strel("line", 5, 4))
```

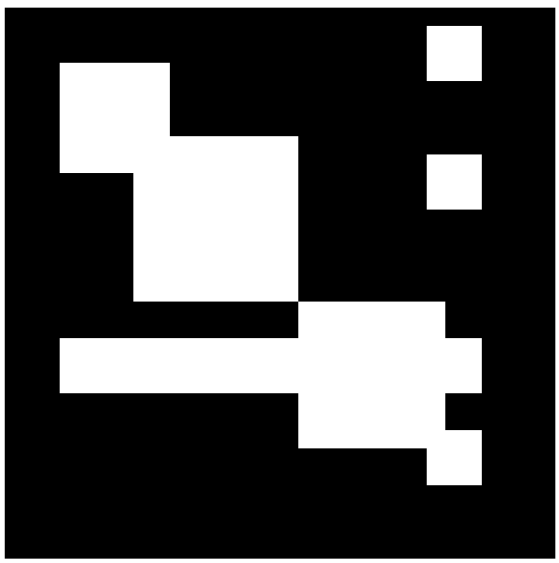
```
ans = 1×5 logical array
   1   1   1   1   1
```

```
se = strel("square", 3);
decompose(se)
```

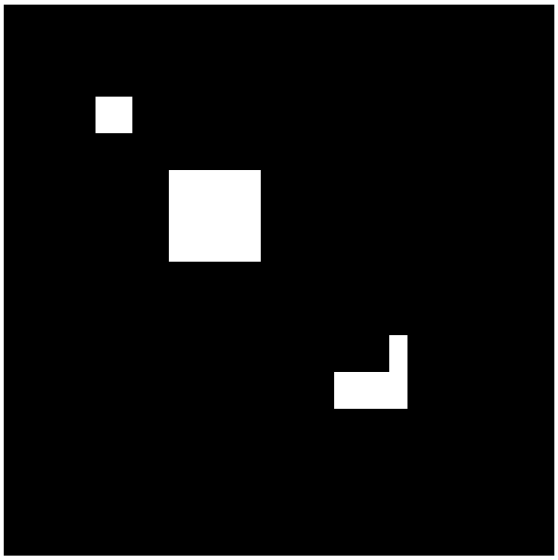
```
ans =
strel is a square shaped structuring element with properties:

      Neighborhood: [3×3 logical]
    Dimensionality: 2
```

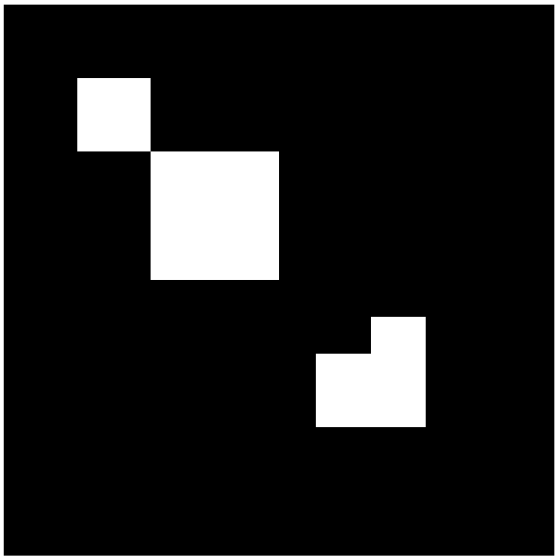
```
dil = imdilate(img, se); imshow(dil)
```



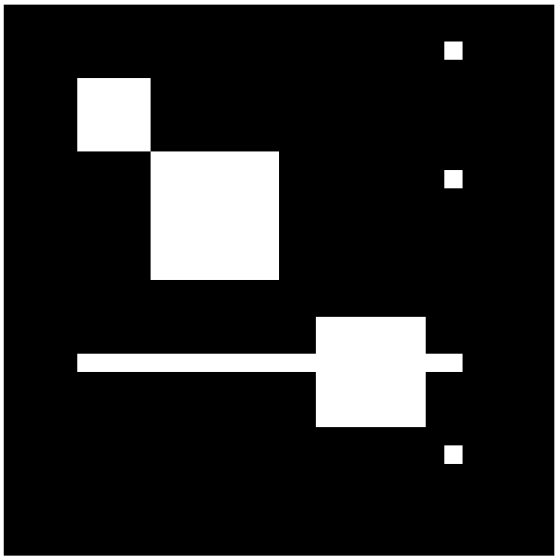
```
erod = imerode(img,se); imshow(erod)
```



```
open = imopen(img,se); imshow(open) % remove small white patches
```



```
close = imclose(img,se); imshow(close) % close small black patches
```



EDGE DETECTION

```
% break_down
I = [10 10 10 10; 10 10 80 80; 10 10 80 80; 10 10 80 80];
imshow(I, [])
```



```
Gx = [1 0; 0 -1];
Gy = [0 1; -1 0];

gx = filter2(Gx, I, 'valid')
```

```
gx = 3×3
     0   -70   -70
     0   -70     0
     0   -70     0
```

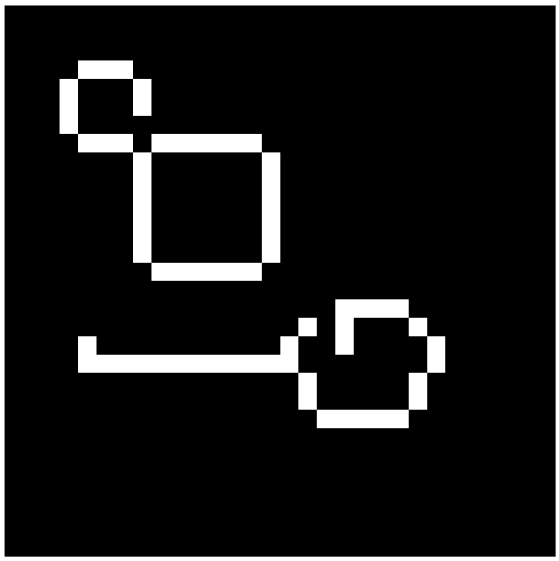
```
gy = filter2(Gy, I, 'valid')
```

```
gy = 3×3
     0     0   -70
     0    70     0
     0    70     0
```

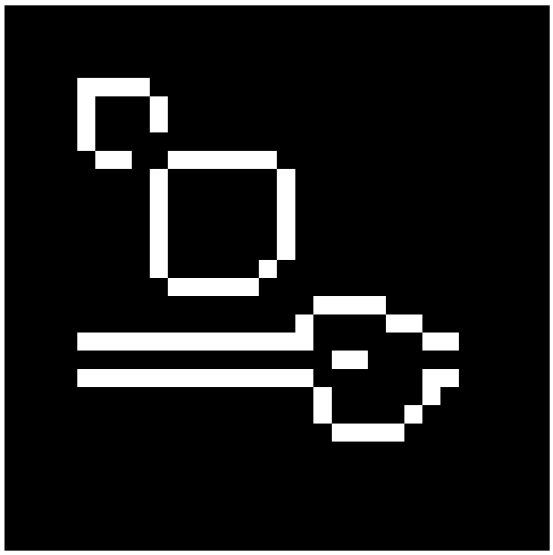
```
G  = sqrt(gx.^2 + gy.^2)
```

```
G = 3×3
         0   70.0000   98.9949
         0   98.9949         0
         0   98.9949         0
```

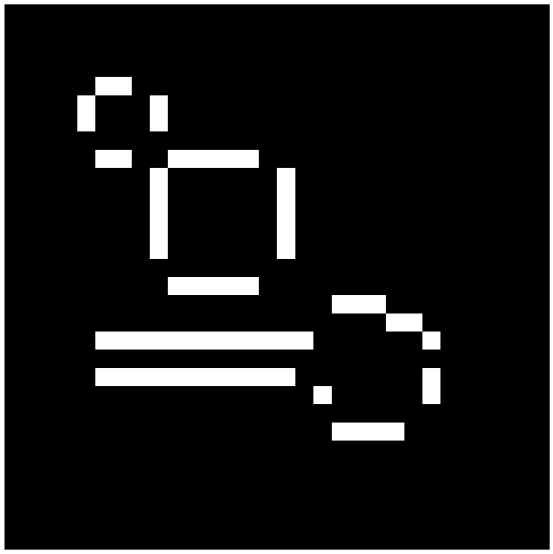
```
% roberts (focuses on cross gradients)
BW = edge(img, 'roberts');
imshow(BW)
```



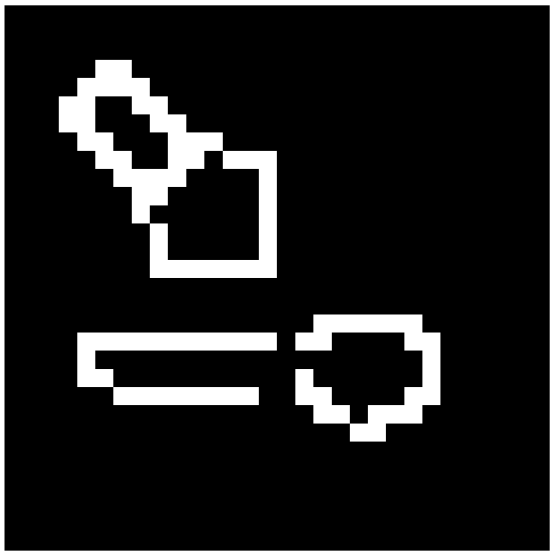
```
% sobel (vertical and horizantal edges)
BW = edge(img, 'sobel');
imshow(BW)
```



```
% prewitt (similar to sobel)
BW = edge(img, 'prewitt');
imshow(BW)
```

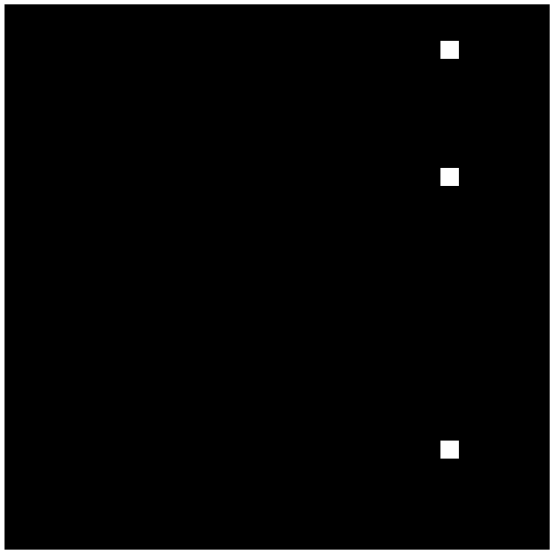


```
% canny
%% smooth >> sobel >> hysteresis
% weak edge kept if it connects strong edge
BW = edge(img, 'canny');
imshow(BW)
```

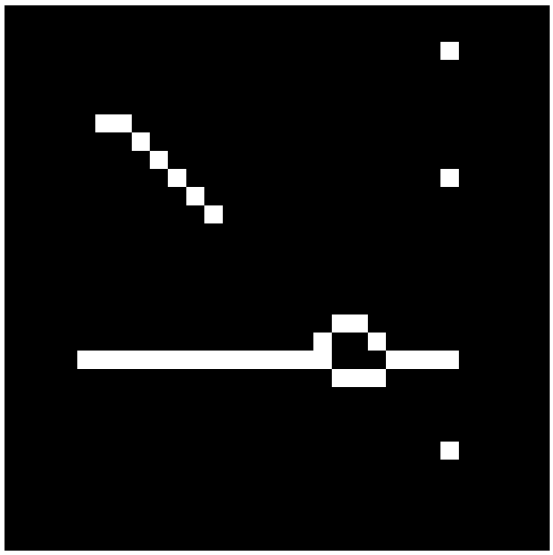


HIT AND MISS

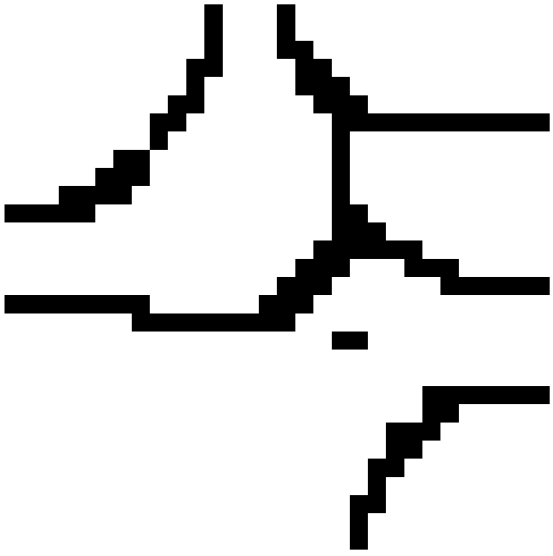
```
SE1 = [0 0 0; 0 1 0; 0 0 0];
SE2 = [1 1 1; 1 0 1; 1 1 1];
iso = bwhitmiss(img, SE1, SE2);
imshow(iso)
```



```
thin = bwmorph(img, 'thin', Inf); imshow(thin)
```

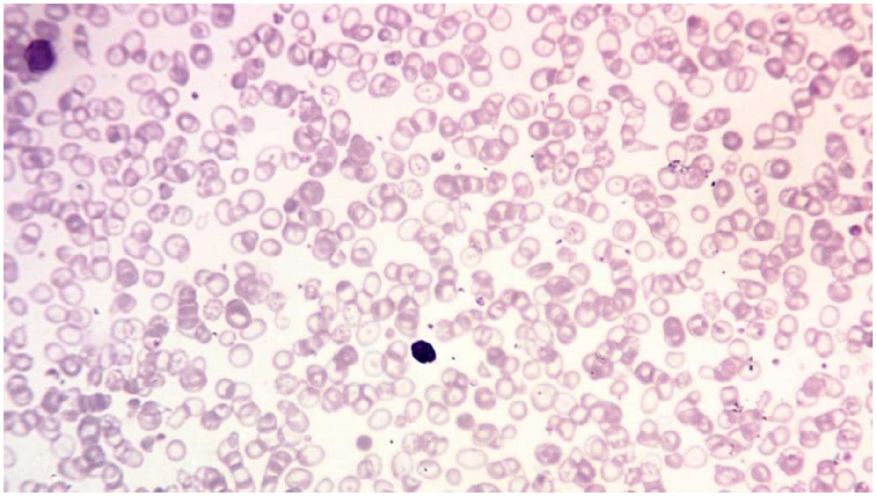


```
thick = bwmorph(img, 'thick', Inf); imshow(thick)
```

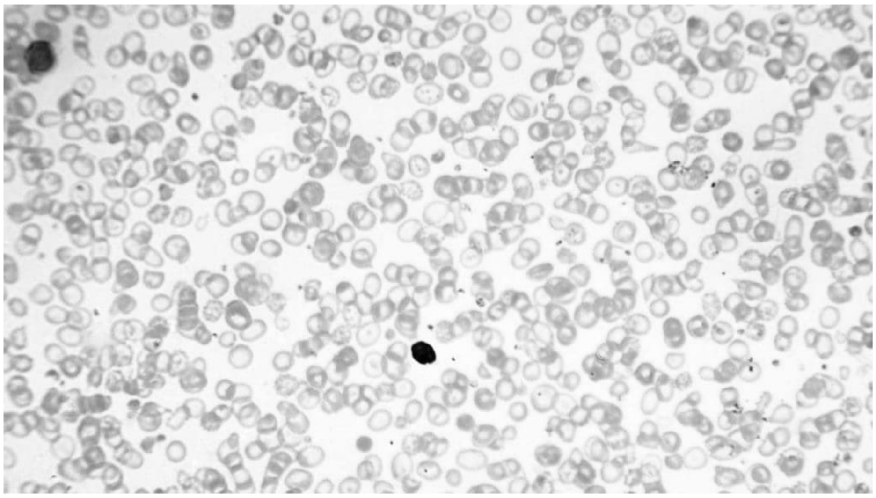


EXAMPLE

```
im = imread("images/rbc.jpg"); imshow(im)
```



```
gray = rgb2gray(im); imshow(gray,[])
```



```
bw = imbinarize(gray, 0.3); imshow(bw)
```